

Chapter

Transpiration

1. The process in which water is lost from the margins of strawberry leaves is:
MAIN 2024

- (a) Osmosis
- (b) Imbibition
- (c) Diffusion
- (d) Guttation

2. Which one of the following helps In the opening of stomata: MAIN 2024

- (a) Cobalt ions
- (b) Potassium ions
- (c) Magnesium ions
- (d) Aluminium ions

3. Permanently open structures seen on the barks of old woody stems :
MAIN 2021

- (a) Stomata
- (b) Hydathodes
- (c) Lenticels
- (d) Epidermal pores

4. Transpiration is useful to the plant because it : **SQP 2021**

- (a) Creates a suction force for absorption of water from the soil
- (b) Helps in Photophosphorylation

- (c) Synthesises glucose
- (d) Splits water molecules.

5. Transpiration :

COMP 2021

- (a) It is the loss of water in the form of droplets from the aerial parts of the plant.
- (b) It is the loss of water in the form of water vapour from the underground parts of the plant.
- (c) It is the loss of water in the form of water vapour from the aerial parts of the plant.
- (d) It is the loss of water in the form of water vapour from all parts of the plant.

6. Cuticle on leaves :

SQP 2021

- (a) Prevents photosynthesis.
- (b) Reduces transpiration.
- (c) Protects leaves from grazing animals.
- (d) Gives colour to leaves.

7. Given below are the advantages of transpiration, except for

SQP 2021

- (a) Cooling
- (b) Suction force
- (c) Distributing water
- (d) Provides oxygen

8. Transpiration decreases with increase in

MAIN 2023

- (a) Temperature
- (b) Wind velocity

(c) Atmospheric pressure

(d) All of the above

9. In a leaf, the cell sap in each cell exerts a outward on the cell wall.

COMP 2023

(a) Wall pressure

(b) Turgor Pressure

(c) Osmotic Pressure

(d) Both (a) and (b)

10. is used as an indicator for a demonstration of transpiration.

MAIN 2021

(a) Litmus paper

(b) Cobalt chloride paper

(c) Both (a) and (b)

(d) None of the above

11. is a device which measures the rate of water intake by a plant.

COMP 2023

(a) Manometer

(b) Thermometer

(c) Potometer

(d) Lactometer

12. The openings on the barks of trees through which transpiration. SQP 2025

13. Small openings present on the bark of woody stems. COMP 2022

14. Three different kinds of structures of a plant from where transpiration occurs. COMP 2023

15. 97% of the total transpiration takes place through____. SQP 2023

16. An instrument used for measuring the rate of transpiration. MAIN 2021

17. The plant in which stomata are absent. MAIN 2021

18. The structure through which guttation takes place. SQP 2023

19. The experiment to demonstrate transpiration. MAIN 2021

20. **Assertion:** Transpiration cools the plant. COMP 2021

Reason: Evaporation of water vapours reduces the temperature of the leaf surface.

(a) If both assertion and reason are true and reason is the correct explanation of assertion.

(b) If both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) If assertion is true but reason is false.

(d) If both assertion and reason are false.

21. **Assertion:** Leaves are reduced to spines in cactus. SQP 2023

Reason: Leaves are modified to form spines to reduce the leaf lamina.

(a) If both assertion and reason are true and reason is the correct explanation of assertion.

(b) If both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) If assertion is true but reason is false.

(d) If both assertion and reason are false.

22. Moist cobalt chloride paper is blue in colour. MAIN 2022

23. The pH of guard cells increase during day time. SQP 2023

24. Guard cells. COMP 2023

25. Stomata MAIN 2022

26. Loss of water from the aerial parts of the plant. SQP 2023

27. How does Transpiration help in the uptake of water from the soil?

SQP 2025

28. Transpiration and Translocation. COMP 2021

29. What is Ganong's Potometer used for? Write any two limitations of this apparatus. SQP 2003

30. How does velocity of wind affect the rate of transpiration? SQP 2023

31. Define transpiration. COMP 2023

32. What is bleeding? MAIN 2021

33. How does the rate of Transpiration differ when there is : SQP 2025

(a) High temperature

(b) High humidity

34. Write the overall chemical equation for Photosynthesis. SQP 2025

35. Mention Two advantages of transpiration. **SQP 2023**

Solution

1. (d) Guttation

Water loss from the leaf margins, as in strawberry leaves, is called **guttation**.

2. (b) Potassium ions

Potassium ions help in the opening of stomata by causing water to enter guard cells, making them turgid.

3. (c) Lenticels

Lenticels are permanently open structures on the bark of old woody stems that allow gas exchange.

4. (a) Creates a suction force for absorption of water from the soil

Transpiration is useful as it **creates a suction force**, helping the plant **absorb water from the soil**.

5. (c) It is the loss of water in the form of water vapour from the aerial parts of the plant.

Transpiration is the **loss of water as water vapour** from the **aerial parts** of a plant.

6. (b) Reduces transpiration

The **cuticle** on leaves **reduces water loss** by limiting transpiration.

7. (d) Provides oxygen

All are advantages of transpiration **except providing oxygen**.

8. (c) Atmospheric pressure

Transpiration **decreases** when **atmospheric pressure increases**.

9. (d) Both (a) and (b)

In a leaf, the **cell sap exerts both wall pressure and turgor pressure** outward on the cell wall.

10. (b) Cobalt chloride paper

Cobalt chloride paper is used as an **indicator to demonstrate transpiration**.

11. (c) Potometer

A **potometer** measures the **rate of water uptake** by a plant.

12. Potometer

13. Lenticels

14. (a) Lenticel

(b) Stomata

(c) Cuticle

15. stomata

16. Potometer

17. Hydrilla (Submerged aquatic plant)

18. Hydathodes

19. Bell jar experiment

20. (a) If both assertion and reason are true and reason is the correct explanation of assertion.

Transpiration cools the plant because the **evaporation of water vapour lowers the leaf temperature**.

21. (a) If both assertion and reason are true and reason is the correct explanation of assertion.

In **cactus**, leaves are reduced to **spines** to **minimise water loss** by reducing the leaf surface area.

22. False, Dry cobalt chloride paper is blue in colour.

23. True

24. Guard cells are located in stomata.

25. **Stomata** : Transpiration and respiration.

26. Transpiration

27. **Transpiration** creates a **suction force** that helps the plant **draw water from the soil** up through the stem.

28.

Transpiration	Translocation
(a) It is the loss of water from the aerial parts of the plant in the form of water vapour.	(a) It is the transportation of food materials within the plant body.
(b) It mainly occurs through the stomata present on the leaves.	(b) It occurs through the vascular tissues — xylem and phloem.

29. Ganong's Potometer is used for measuring the rate of transpiration in plants. The apparatus has some limitations such as difficulty in introducing an air bubble in the capillary tube, and the plant also uses some water for photosynthesis and other metabolic activities which cannot be measured.

30. When the velocity of wind increases, the rate of transpiration also increases because the moving air removes water vapour from around the leaf surface, preventing saturation. When the wind velocity is low, water vapour remains near the leaf surface, reducing the rate of transpiration.

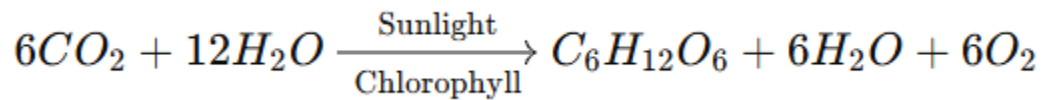
31. Transpiration is the loss of water in the form of water vapour from the aerial parts of a plant.

32. Bleeding is the exudation of plant sap through a cut or injured part of a plant.

33. (a) At high temperature, the rate of transpiration increases.

(b) At high humidity, the rate of transpiration decreases.

34.



35. (i) Transpiration helps in cooling the plant and maintaining its temperature. When water evaporates from the leaves, it absorbs heat energy, thereby reducing the plant's temperature.

(ii) Transpiration produces a suction force that enables the upward movement of water and minerals from the roots to different parts of the plant, aiding in its growth and nourishment.